

DRONE BASIS VECTORS

Linear Algebra 1

1 BASIS VECTORS

→ $\hat{i} = (1,0,0)$
x-direction

→ $\hat{j} = (0,1,0)$
y-direction

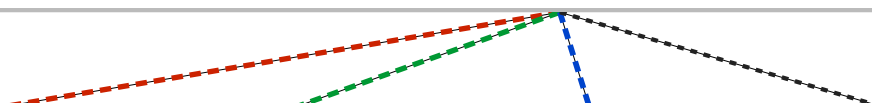
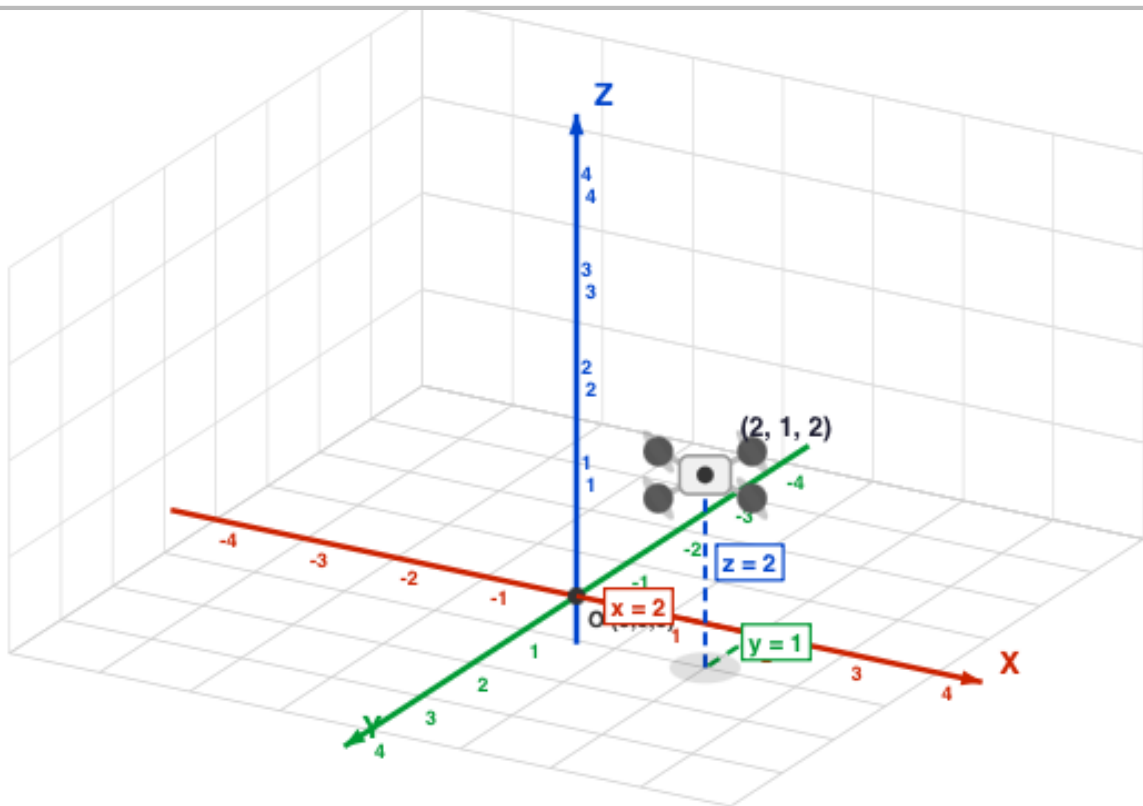
← $\hat{k} = (0,0,1)$
z-direction

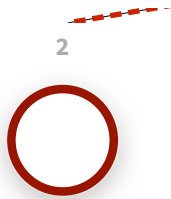
2 LINEAR COMBINATION

$$\vec{v} = x\hat{i} + y\hat{j} + z\hat{k}$$

Any position or movement in 3D space can be written as a combination of the three basis vectors.

x, y, z are scalar values.





CONTROL X

(Left – Right)



CONTROL Y

(Forward – Backward)



CONTROL Z

(Up – Down)



MANUAL SUPPORT

(Move by Stick / Failure

Demo)

Drone Position

(2, 1, 2)

3 HOW IT WORKS

- Each control string represents motion along one axis.
- Pull a string to tilt or move the drone in that direction.
- If one motor fails, balance is lost.

4 MOTOR FAILURE EXAMPLE

Disconnect one string to simulate a failed motor.

